

Task-Conditioned Cognitive Architecture Co-Design for LLM Agents: A Controlled Study of Generalization, Cost, and Failure Modes

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Abstract

LLM agents are usually deployed with a fixed, manually designed cognitive architecture. We ask whether per-task co-design of an agent’s reasoning strategy, episodic memory/exemplar policy, and context allocation—above a frozen backbone—materially changes capability–cost behavior, and whether task-conditioned search generalizes. We present NeuroEvoScientist, a controlled study over a structured family of agent configurations with leakage-safe calibration splits, raw multi-objective logging, task-aware phenotype canonicalization, pre-registered holdout evaluation, and paired item-level statistics across two frozen backbones (Qwen2.5-1.5B and 7B). On untouched GSM8K holdout, a searched configuration outperforms the strongest fixed baseline by +29.0 percentage points at 1.5B and +16.7 points at 7B (McNemar $p < 0.001$), while controlled same-genome ablation shows memory contributions that grow with backbone scale. Search-side architecture preferences differ clearly and stably by task—a long-context scientific-QA task shifts the optimum to retrieval-based memory, as pre-registered—yet dev-selected task-specific architectures do not universally generalize: on PubMedQA and QASPER holdout they are beaten by the configuration selected for GSM8K. A QASPER configuration that works at 1.5B collapses at 7B; our diagnostic shows this is an answer-format/extraction interaction, not a capability failure. Equal-budget evolutionary search does not beat random search in this compact space, and a real trainable Mamba-2 substrate is not selected as beneficial; we report both as audit findings. The work contributes a leakage-safe, falsifiable protocol for agent architecture co-design and a careful map of where task-conditioned specialization does—and

does not—generalize.

1 Introduction

LLM agents are typically assembled by hand: a reasoning pattern, a memory mechanism, and a context budget are chosen once and reused across tasks. Yet different tasks place different demands on each component—multi-step arithmetic rewards explicit reasoning steps, biomedical judgment rewards calibrated abstention, and long-document question answering rewards context allocation and evidence retrieval. We ask two questions. First, do these cognitive architecture choices materially change an agent’s capability–cost behavior when the underlying language model is kept frozen? Second, can automatic, task-conditioned search over such choices find configurations that remain superior on data untouched by the search itself?

We answer both with NeuroEvoScientist, a controlled co-design framework in which an agent configuration is a structured genome over reasoning strategy, episodic memory policy, and context policy, evolved with multi-objective selection and evaluated under a leakage-safe, pre-registered protocol. Our study deliberately audits its own search procedure: we enumerate the compact search space exhaustively to obtain a ground-truth landscape, we compare evolutionary search to equal-budget random search at budgets below full coverage, and we hold out benchmark items that the search never sees.

Our findings are deliberately two-sided. On the positive side, architecture choices matter enormously: searched configurations beat the strongest fixed baseline by up to +29 percentage points on held-out GSM8K, task-specific gene distributions are stable across seeds, and a pre-registered prediction—that

a long-context task shifts the optimum toward retrieval memory—is confirmed. On the negative side, dev-selected task-specific architectures do not universally generalize; equal-budget random search matches evolutionary search in our compact space; and a real Mamba-2 substrate, though correctly implemented and trainable, is not selected as beneficial. We report these negative results as part of the contribution.

2 Related Work

Agent architectures and workflow optimization. LLM agents commonly chain reasoning and acting through manually designed scaffolds (Yao et al., 2023; Schick et al., 2023) or self-reflection loops (Shinn et al., 2023); DSPy compiles declarative LLM calls into optimizable pipelines (Khatab et al., 2023). These frameworks optimize prompts or workflow structure; we instead co-design an agent’s cognitive configuration—reasoning strategy, episodic memory policy, and context allocation—under explicit capability–cost objectives with a frozen backbone.

Automated agent search. Most closely related are automated agent-design frameworks: ADAS searches over agent programs expressed in code (Hu et al., 2024), and AgentSquare searches a modular design space of planning, reasoning, tool-use, and memory modules (Shang et al., 2025). Both optimize for performance alone. We differ by (i) searching per-task cognitive configurations with real memory substrates rather than workflow topology; (ii) evaluating under multiple raw objectives instead of a single score; and (iii) pre-registered held-out auditing with paired statistics, reporting where specialization does and does not generalize.

Prompt and reasoning strategy optimization. Chain-of-thought prompting materially affects reasoning accuracy (Wei et al., 2022), and EvoPrompt evolves prompt text (Guo et al., 2023). We treat the reasoning strategy as one gene jointly searched with memory and context policy.

Episodic and retrieval memory. Retrieval-augmented generation grounds LLMs in external evidence (Lewis et al., 2020). Our memory

policies manage a calibration-only experience bank, and we quantify when episodic memory helps, hurts, or is neutral across tasks and backbones.

Evolutionary computation and NAS. Neural architecture search is surveyed by Elsken et al. (2019); regularized evolution is a strong NAS baseline (Real et al., 2019), and multi-objective selection follows NSGA-II-style non-dominated sorting with crowding distance (Deb et al., 2002). Bayesian optimization is the standard sample-efficiency reference (Shahriari et al., 2016). Unlike NAS over backbone weights, our search operates above a frozen LLM, and we audit the search procedure against equal-budget random search on a fully enumerated landscape.

Automated discovery and memory models. AlphaEvolve evolves programs through LLM proposals and evaluators (Google DeepMind, 2025); we share the evolutionary loop but change the object of evolution. Mamba and Mamba-2 provide the selective state-space models from which our real trainable memory substrate is built (Gu and Dao, 2023; Dao and Gu, 2024).

Benchmarks and backbones. We evaluate on GSM8K (Cobbe et al., 2021), PubMedQA (Jin et al., 2019), and QASPER in its LongBench form (Dasigi et al., 2021; Bai et al., 2023), using frozen Qwen2.5-Instruct backbones (Yang et al., 2024). We claim no priority over any of these works; our contribution is a controlled, leakage-safe, multi-objective audit of task-conditioned co-design, including its failure modes. Parameter-efficient adapters such as LoRA (Hu et al., 2021) are related as adapter technology only; no LoRA gene is searched or claimed in this paper.

3 Problem Formulation

An agent configuration is a genome over three gene families: memory policy, reasoning strategy, and context policy. Given a task and a frozen backbone, we seek configurations that occupy favorable positions on the capability–cost Pareto frontier—not a single scalar optimum. Task conditioning means the preferred configuration may differ per task; the scientific question is whether search-side prefer-

ences correspond to generalizable holdout advantages or merely to search-split overfitting.

4 Method

4.1 Two protocol layers

Compact audit layer. A flat genome over memory {recency, retrieval, mamba2, hybrid}, reasoning {direct, cot, verify, planner}, and context policy {full, truncated, answer_only} defines 48 configurations. We evaluate all of them per benchmark to obtain an exhaustive landscape with raw objectives—capability, efficiency, prompt-token cost, latency, and adaptation cost—used as an analysis oracle for the search-efficiency audit.

Structured co-design layer. The final search space refines each gene with conditional sub-genes: memory recall width and similarity metric (hashed bag-of-words cosine; explicitly not a learned dense retriever), Mamba-2 state size, hybrid mixing fraction, reasoning depth and verifier passes, exemplar word budget and count, input-context word budget for long-document tasks, and a per-candidate substrate-adaptation budget. Normalization removes semantically inactive sub-genes before hashing, so duplicate phenotypes are never double-counted; task-aware canonicalization additionally drops the input-context budget outside long-context tasks.

4.2 Genes are real mechanisms

All memory banks are built exclusively from calibration splits; test items never enter the bank. recency returns the most recent k entries; retrieval returns similarity top- k ; mamba2 runs a real, trainable Mamba-2 substrate over the bank as an ordered sequence and scores candidates against the resulting order-dependent state; hybrid mixes retrieval with the most recent entry. Reasoning genes are distinct instruction templates; context genes change real exemplar verbosity and document allocation. Candidates carrying the trainable substrate receive a fixed adaptation budget (48 calibration samples, 30 AdamW steps, fixed learning rate and seed) with the backbone frozen; substrate weights can be inherited from parents as an appendix mechanism evaluated separately.

4.3 Operators

Mutations are local (neighboring values for ordered sub-genes), crossover swaps semantic blocks and normalizes, and selection is NSGA-II-style non-dominated sorting with crowding-distance diversity over raw objectives; scalar fitness is used only for logging.

5 Experimental Setup

5.1 Benchmarks and splits

Benchmark	dev/search	holdout	calibration
GSM8K	test[0:100]	test[100:1319]	train split
PubMedQA	samples[0:100]	samples[100:500]	samples[500:1000]
QASPER	items[0:50]	items[50:150]	items[150:200]

Table 1: Leakage-safe splits. Pairwise disjointness is enforced by unit tests; selection locks were committed before any holdout inference.

5.2 Backbones and decoding

Qwen2.5-1.5B-Instruct for search and selection; Qwen2.5-7B-Instruct for transfer only (no re-search). Greedy decoding, 256 max new tokens, batch 32, FP16. Exact model revisions and dataset acquisition details are in the reproducibility statement (Section 11).

5.3 Audit protocols

The compact-space audit compares online evolutionary search and uniform random search at evaluation budgets {12, 24, 36, 48} across 20 search seeds (best capability, Pareto hypervolume, regret, epsilon-Pareto hits, evaluations-to-threshold, AUC). All key holdout comparisons are paired at the item level: bootstrap 95% CIs (10,000 resamples, fixed seed) and exact McNemar tests.

6 Main Results

6.1 R1: architecture choices matter materially

On untouched GSM8K holdout, A_gsm beats the strongest fixed baseline by +29.0 percentage points at 1.5B and +16.7 at 7B (McNemar $p < 0.001$). On PubMedQA, searched configurations produce competitive capability-cost tradeoffs but not universal capability superiority—a fixed retrieval baseline remains ahead at 1.5B—and we report this asymmetry rather than hiding it.

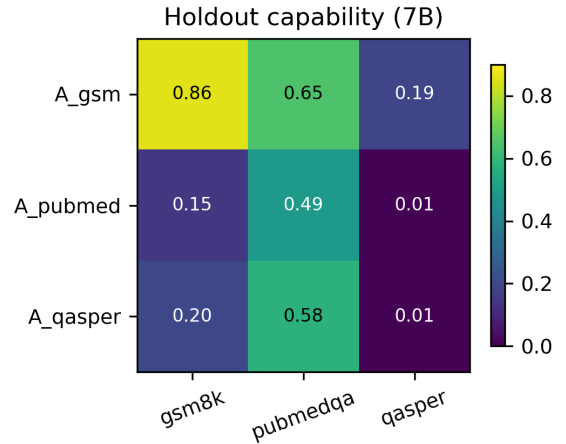
Config	GSM8K		PubMedQA		QASPER (F1)	
	1.5B	7B	1.5B	7B	1.5B	7B
A_gsm (searched)	0.519	0.860	0.490	0.648	0.188	0.188
A_pubmed (searched)	0.097	0.147	0.373	0.488	0.111	0.006 [†]
A_qasper (searched)	0.208	0.196	0.563	0.578	0.138	0.006 [†]
best fixed baseline	0.229	0.693	0.598	0.603	0.176	0.015
no_memory (ablation)	0.519	0.734	0.363	0.578	0.201	0.214

Table 2: Holdout capability (accuracy; QASPER: LongBench-compatible normalized QA F1 on extracted answer) for the three task-selected configurations, strongest fixed baselines, and the Phase-20 no-memory reference (the A_gsm genome with memory disabled). Controlled causal memory ablations are reported separately in Table 4. [†]Marker-side extraction mismatch diagnosed in Section 7 (not a capability collapse).

Final-generation gene distribution per task (3 seeds)

	gsm8k	pubmedqa	qasper
memory_type	none (69%)	none (42%)	none (44%)
reasoning	cot (52%)	verify (42%)	direct (79%)
context_mode	full (54%)	answer_only (40%)	answer_only (42%)
exemplar_count	0 (23%)	1 (44%)	0 (31%)
input_context_budget	2048 (33%)	4096 (54%)	512 (46%)

Figure 1: Final-generation gene distribution per task (mode and share over three dev-search seeds).



6.2 R2: task-conditioned selection is stable and interpretable

Across three search seeds, GSM8K converges to chain-of-thought with light recency memory, PubMedQA converges bit-identically to no-memory direct answering on all three seeds, and QASPER converges to retrieval-based memory with many exemplars and a reduced document budget—matching our pre-registered prediction that a long-context task shifts the optimum toward retrieval (Figure 1). A sensitivity audit after task-aware phenotype canonicalization leaves these families unchanged.

6.3 R3: dev-selected preferences do not universally generalize

On GSM8K holdout, the own-task configuration dominates every alternative by +31 to +71 points ($p < 0.001$) in cross-task paired comparisons. On PubMedQA and QASPER holdout, however, the task-selected configurations are beaten—significantly—by the GSM8K-selected configuration (Table 3). Search-side specialization is real, but dev-slice preferences can overfit.

Figure 2: Holdout capability transfer matrix at 7B: rows are task-selected configurations, columns are holdout benchmarks.

6.4 R4: memory contribution is controlled and scales with backbone

Under same-genome ablation, removing episodic memory is neutral at 1.5B on GSM8K but removes 12.6 points at 7B ($p < 0.001$); on PubMedQA, enabling memory on the frozen configuration adds +18.0 points at 1.5B; on QASPER the effect is small (+1.5 points; Figure 3). Memory causality is claimed only where the comparison is controlled.

7 Generalization Audit and Failure Analysis

7.1 Search-efficiency audit

The compact-space audit finds no search-efficiency advantage for evolutionary search over equal-budget random search at any budget or benchmark (differences below 0.01 in AUC and hypervolume across 20

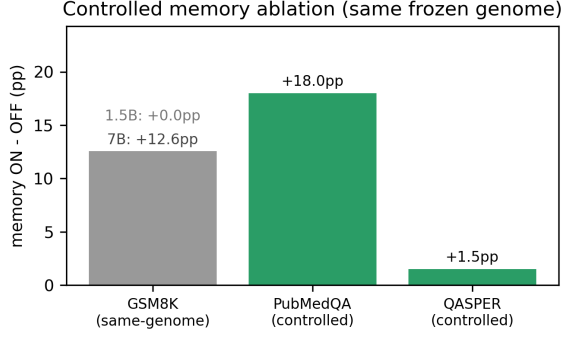


Figure 3: Controlled same-genome memory ablation across backbones.

Holdout task	Comparison (own vs cross)	1.5B
GSM8K	A_gsm vs A_pubmed	+42.2***
	A_gsm vs A_qasper	+31.0***
PubMedQA	A_pubmed vs A_gsm	-11.8***
	A_pubmed vs A_qasper	-19.0***
QASPER	A_qasper vs A_gsm	-5.0*
	A_qasper vs A_pubmed	+2.7*

Table 3: Paired cross-task differences in holdout capability (percentage points; bootstrap CI + exact McNemar; * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$). Positive favors the task-selected configuration.

seeds; Figure 4). We therefore make no optimizer-superiority claim; evolution is the architecture-generation mechanism, and our contribution is the task-conditioned co-design evidence and the audit protocol itself.

7.2 Negative results kept

A real, trainable Mamba-2 substrate participates in the search but is not selected as beneficial under our protocol; we keep this negative result. Weight inheritance improves post-adaptation loss in all 20 paired cases but yields no capability difference in any of them; it is reported as an appendix mechanism.

7.3 QASPER-7B failure diagnosed

Several QASPER configurations appear to collapse at 7B (near-zero extracted-answer F1). Our frozen diagnostic shows marker compliance is perfect (100%) while extracted-answer F1 is zero and whole-continuation F1 remains positive—the 7B backbone places its answer before or at the marker instead of after it (Figure 5). This is a backbone-prompt/extraction interaction, not a capability failure.

Task	Pair (same genome)	Backbone	ON	OFF
GSM8K	A_gsm $\pm$ memory	1.5B	0.519	0.519
		7B	0.859	0.734
PubMedQA	frozen recency $\pm$ memory	1.5B	0.553	0.373
QASPER	A_qasper $\pm$ memory	1.5B	0.125	0.111

Table 4: Controlled same-genome memory ablation (capability; the only causal memory comparisons in this paper).

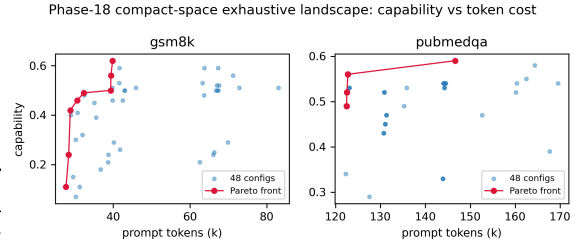


Figure 4: Phase-18 compact-space exhaustive landscape (48 configurations per benchmark): capability vs prompt-token cost, with the global Pareto front.

8 Mechanism Analysis

Fourteen auditable case studies link genome choices to inference behavior: successful and failed GSM8K items with recalled exemplars, PubMedQA decisions, and paired QASPER 1.5B/7B failures with raw outputs and extracted answers. These cases illustrate how exemplar content, context budget, and answer-formatting instructions jointly shape outcomes.

9 Limitations

Our study uses a single model family (Qwen2.5 at 1.5B and 7B), a compact audited space, prompt-level reasoning genes, and a single adaptation protocol. PubMedQA’s three-way ceiling limits capability separation, and QASPER’s answer extraction is marker-sensitive, which we diagnose rather than hide. Cross-backbone latency is not compared without normalization. Evolutionary superiority is not supported in our space; we do not claim it. Small dev slices can overfit, as our own PubMedQA/QASPER results show.

10 Conclusion

Cognitive architecture choices materially change agent capability-cost behavior. Automatic co-design exposes clear, stable, in-

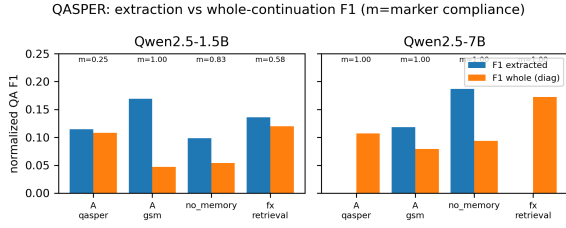


Figure 5: QASPER diagnostic: extracted-answer vs whole-continuation F1 at 1.5B and 7B, with marker compliance. The 7B “collapse” is an answer-format/extraction interaction.

Config	Backbone	F1 (extr.)	F1 (whole)
A_qasper	7B	0.000	0.107
fixed_retrieval	7B	0.000	0.172
A_gsm	7B	0.118	0.079
no_memory	7B	0.187	0.094
A_qasper	1.5B	0.115	0.108
A_gsm	1.5B	0.169	0.047

Table 5: QASPER diagnostic (12 fixed holdout items, frozen protocol): marker compliance is perfect at 7B while extracted-answer F1 collapses, diagnosing an answer-format/extraction interaction rather than a capability failure.

interpretable task-conditioned preferences—and a rigorous held-out audit is what distinguishes search-side specialization from generalizable advantage. We release the full protocol, landscapes, per-item predictions, statistics, and claim audit to make this kind of study falsifiable by construction.

11 Reproducibility and Ethics Statement

All experiments are frozen at the locked commits; exact model revisions, dataset acquisition details, protocol parameters, cache semantics, and the result-to-evidence map are documented in the anonymous reproducibility package (supplementary material). The work uses public benchmarks and public models, involves no human subjects, and reports negative results and failure modes explicitly.

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